A Constructive Introduction to Quantum Field Theory

Φ^4_3 on the torus II: a priori estimates

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SISSA lectures, Lecture 6 (provisional) — notes: Lecture 6, §§13–15

Where we are

At each lattice N, exactly (Lecture 5):

$$\mathrm{d}Y_q = [b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + H_q(Y_q)\,\mathrm{d}W_q,$$
$$\mathrm{d}Z_q = -[b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + \Xi_q\,\mathrm{d}W_q,$$

$$\psi = Y + Z = \int \mathrm{Cov}\,\mathrm{d}W.$$

Inputs.

A) Coercivity: $H_q(y) \asymp (K_q + \gamma y^2)^{-1}$, $\|H_q\|_{p\to p} \le (1+\varepsilon_q)/q$, $\varepsilon_q = C(g_q + g_q^2)$.

B) Source: $\|b_q^{\mathrm{b}}\|_\infty \le C\gamma^{7/2} q^{-5/2}$, $\|b_q^{\mathrm{I}}(y)\|_2^2 \le C\gamma^4 q^{-5} \|t_q(y)\|_6^6$.

C) Coordinates: $K_q(Y_q + \varrho_q(Y_q)) + \frac{\gamma}{3} Y_q^3 = P_q$, $\|D\,\varrho_q\|_{p\to p} \le \varepsilon_q$.

Goal

Z small and Y rarely at large fields, **uniformly in N**. Neither equation can be estimated alone.

How FBSDE estimates usually work, and why not here

Usual (Ma–Yong): Lipschitz coefficients, Itô on $\|Z\|^2$ and Gronwall; close on a short interval or under monotonicity.

Here: infinite interval, coefficients not uniformly Lipschitz, the coupling is not small. Instead we use **structure**:

- the backward equation is **linear in Z** given Y: a conditional representation with an explicit propagator;
- the forward equation is, after subtracting the Gaussian part, a cubic **elliptic** equation at each resolution: coercive, estimated **pathwise**;
- the two estimates are linked by two weighted energies with **matched weights**.

At fixed N, T all moments are finite (finite dimensions): used only to justify absorptions. Ma–Yong 1999

The energies

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Subtracting the Gaussian part

Free field at resolution q driven by the **same** innovation:

$$X_q = \int_{q_0}^{q} C_s \, \mathrm{d}W_s \sim \mu^{Q_q}, \qquad K_q X_q = W_q + \int_{q_0}^{q} X_s \mathrm{d}s.$$

Also $P_q = W_q + \int (Y + Z)$. With $V_q = Y_q + \varrho_q(Y_q) - X_q$ and $F_q^Z = \int_{q_0}^{q} Z_s \mathrm{d}s$, the noise cancels:

$$\boxed{K_q V_q + \frac{\gamma}{3} Y_q^3 = \int_{q_0}^{q} V_s \, \mathrm{d}s - \int_{q_0}^{q} \varrho_s(Y_s) \, \mathrm{d}s + F_q^Z, \qquad Y_q = V_q + X_q - \varrho_q(Y_q).}$$

- At each q: $(q - \Delta)\, V + \frac{\gamma}{3} (V + X - \varrho)^3 = f$, a nonlinear Helmholtz equation; X is a shift inside the cube.
- The memory has **coefficient one**: $\int_{q_0}^{q} V \approx q\, V$. It must be absorbed, not treated as forcing.
- The backward equation enters only through $F^Z = \int Z$. Pathwise.

Thresholds and energies

$$w_q = \operatorname{sgn}(V_q)\left(|V_q| - \tfrac{3}{4} r_q\right)_+, \qquad t_q = Y_q 1_{|Y_q| > r_q}, \qquad \xi_q = X_q 1_{|X_q| > r_q/32}.$$

$$E_T = \frac{1}{\gamma}\int q^{1/2}\|Z_q\|_2^2 \,\mathrm{d}q \qquad \text{backward energy}$$

$$J_T = \int q^{-3/2}\left[q\,\|w_q\|_4^4 + \gamma\|w_q\|_6^6 + \gamma\|t_q\|_6^6\right]\mathrm{d}q \qquad \text{large-field action}$$

$$G_T = \int \left[q^{-1/2}\|\xi_q\|_4^4 + \gamma\, q^{-3/2}\|\xi_q\|_6^6\right]\mathrm{d}q \qquad \text{Gaussian tails}$$

quantity	weight	reason
$\|Z_q\|^2$	$q^{1/2}$	Hardy: $\int q^{-3/2}\|F^Z\|^2 \le 16\int q^{1/2}\|Z\|^2$
$\gamma\|t_q\|_6^6$	$q^{-3/2}$	$q^{7/2}\|b^{\mathrm{I}}\|^2 \le C\gamma^3\cdot\gamma\, q^{-3/2}\|t\|_6^6$
$\|b_q^{\mathrm{I}}\|^2$	$q^{7/2}$	propagator s/q, $\int_q^\infty s^{-3/2} = 2q^{-1/2}$

Only **thresholded** sixth powers: unthresholded, $\mathbb{E}\|X_q\|_6^6 \asymp q^{3/2}$ for $2q_0 \le q \lesssim N^2$, so $\int \gamma q^{-3/2}\,\mathbb{E}\|X_q\|_6^6 \approx \gamma N^2$ is not uniform in N.

Two Hardy inequalities

Lemma. *For f on $[q_0,\infty)$, extended by zero to $(0,q_0)$,*

$$\left|q^{-1}\int_0^q f\right|_{L^4(q^{-1/2}\mathrm{d}q)} \le \frac{8}{7}\|f\|_{L^4(q^{-1/2}\mathrm{d}q)}, \qquad \int q^{-3/2}\left|\int_0^q f\right|^2 \le 16\int q^{1/2}\|f\|^2.$$

Proof: $q^{-1}\int_0^q f = \int_0^1 f_{q\theta}\mathrm{d}\theta$, Minkowski; $\int_0^1 \theta^{-1/8} = \frac{8}{7}$, $(\int_0^1 \theta^{-3/4})^2 = 16$.

The problem

$\frac{8}{7} > 1$: the mass term alone (coefficient one) does not absorb the memory. The missing margin comes from the **cube**: at large fields it adds $\frac{1}{4}q|w|^4$, and $1+\frac{1}{4}-\frac{8}{7} = \frac{3}{28} > 0$.

Forward coercivity

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The scalar model

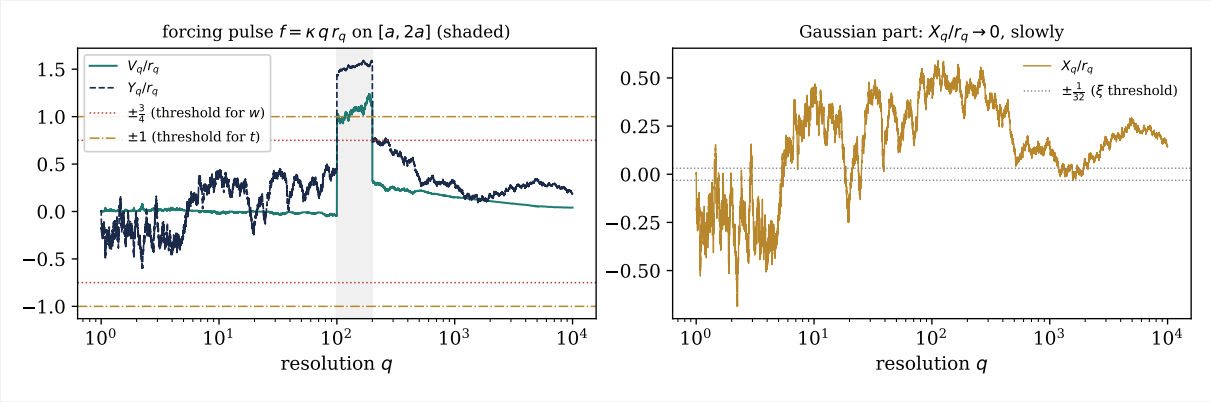
Lemma. (scalar forward coercivity) *If* $q\,v_q + \frac{\gamma}{3}(v_q + X_q)^3 = \int_{q_0}^{q} v_s\,\mathrm{d}s + f_q$ *on* $[q_0, T]$, *then*

$$\frac{3}{28}\int q^{-1/2} w^4 + \frac{\gamma}{16}\int q^{-3/2} w^6 \le \frac{4225}{6}\gamma \int q^{-3/2}\xi^6 + \frac{12}{\gamma}\int q^{-3/2} f^2,$$

and $|t_q| \le \frac{32}{7}(|w_q| + |\xi_q|)$. *Pathwise, uniform in* γ.

The scalar model: a simulation

Forcing pulse $f_q = \kappa\, q\, r_q\, 1_{[a,2a]}(q)$, κ of order one. Thresholds: $\pm\frac{3}{4}$ for V_q/r_q, ± 1 for Y_q/r_q; the Gaussian part X_q/r_q on its own panel.



The pulse pushes V past $\frac{3}{4}r$; the cube caps the excursion and brings it back. On this range $|X_q| > r_q/32$ most of the time: ξ_q is not rare until g_q is small (here $q \gg 10^6$). In this realisation the Gaussian part stays below the threshold, $|X_q|/r_q < \frac{3}{4}$.

Proof: test with w^3

Where $w \neq 0$: $|v| = |w| + \frac{3}{4} r$, same sign.

- **Mass:** $q\, v\, w^3 = q|w|^4 + \frac{3}{4} q\, r |w|^3$.
- **Cube:** $|X| \leq \frac{r}{32} + |\xi|$, so $\operatorname{sgn}(w)\,(v+X) \geq |w| + \frac{11}{16} r - |\xi|$; the cube is increasing, and

$$\frac{\gamma}{3} w^3 (v+X)^3 \geq \frac{\gamma}{12} w^6 + \frac{q}{4} w^4 - \frac{4225}{6} \gamma\, \xi^6 \qquad (\gamma r^2 = q).$$

Large-field coercivity: a sixth power and a quarter of the mass.

- **Memory:** $|v - w| \leq \frac{3}{4} r$, $\int_{q_0}^{q} r \leq \frac{2}{3} q r$: $w^3 \int v \leq \frac{1}{2} q\, r |w|^3 + w^3 \int w$; the first is absorbed by $\frac{3}{4} q\, r |w|^3$.
- **Forcing:** $w^3 f \leq \frac{\gamma}{48} w^6 + \frac{12}{\gamma} f^2$.

Multiply by $q^{-3/2}$, integrate, Hölder and Hardy: $\int q^{-3/2} w^3 \int w \leq \frac{8}{7} \int q^{-1/2} w^4$, and $\frac{5}{4} - \frac{8}{7} = \frac{3}{28}$.

Two elementary inequalities

For $x, \rho \geq 0$: $\frac{1}{4}x^3 + 64\rho^3 - 3x^2\rho = \frac{1}{4}(x-8\rho)^2(x+4\rho) \geq 0$, hence

$$(x-\rho)^3 \geq \frac{3}{4}x^3 - 65\rho^3.$$

With $x = u + \frac{11}{16}r$: $u^3x^3 \geq u^6 + 3(\frac{11}{16})^2 r^2 u^4$, $\frac{3}{4}(\frac{11}{16})^2 = \frac{363}{1024} > \frac{1}{4}$, and $\frac{65}{3}u^3\rho^3 \leq \frac{1}{6}u^6 + \frac{4225}{6}\rho^6$. So

$$\frac{1}{3}u^3\left(u + \frac{11}{16}r - \rho\right)^3 \geq \frac{1}{12}u^6 + \frac{1}{4}r^2u^4 - \frac{4225}{6}\rho^6.$$

What the scalar lemma says

The excess of the forecast over the threshold is controlled by how often the **Gaussian** field is large (known, N-independent) and by the **backward** forcing. No smallness of the coupling; pathwise.

The lattice version

Proposition. (forward coercivity) *For q_0 large, at every N, T, pathwise:* $J_T \le C\,(G_T + E_T)$, $\quad \|G_T\|_{L^p} \le C\gamma^3 p^5$.

New terms:

- **Laplacian:** $w^3 = \phi'(V)$, ϕ convex, so $\langle \phi'(V), -\Delta V\rangle = N^{-1}\sum_{x\sim x'}(\phi'(V_x) - \phi'(V_{x'}))\,(V_x - V_{x'}) \ge 0$.
- **Coordinate correction:** $\varrho = \varrho^0 + \varrho^1$, $\|\varrho^0\|_\infty \le \varepsilon\, r$ (threshold shift; memory absorbed by the spare $\frac{1}{4} q\, r|w|^3$), $\|\varrho^1\|_j \le C\,\varepsilon\,(\|w\|_j + \|\xi\|_j)$ (memory absorbed by $\frac{3}{28}$). Operator bounds: ϱ is nonlocal but small.
- **Backward forcing:** $\gamma^{-1}\int q^{-3/2}\|F^Z\|^2 \le 16\, E_T$.
- **Gaussian tails:** $\mathrm{Var} X_q(x) \lesssim \sqrt{q}$ vs threshold $\frac{1}{32}\,(q/\gamma)^{1/2}$: $|\|\xi_q\|_j^j|_p \le C_j p^{j/2} q^{j/4} \mathrm{e}^{-c\sqrt{q}/(p\gamma)}$.

X is Gaussian under the interacting measure: **no information on the interacting field is used**.

Backward feedback and closure

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The backward representation

Propagator: $\partial_s \mathcal{P}_{q,s} = \mathcal{P}_{q,s} H_s(Y_s)$, $\mathcal{P}_{q,q} = I$, and by (A)

$$\|\mathcal{P}_{q,s}\| \le L_0 \frac{s}{q}, \qquad L_0 = \exp\left(\int_{q_0}^{\infty} \varepsilon_u \frac{\mathrm{d}u}{u}\right) \le \mathrm{e}^{C(g_{q_0} + g_{q_0}^2)}.$$

Lemma. (conditional representation)

$$Z_q = \mathbb{E}_q\left[\mathcal{P}_{q,T} Z_T + \int_q^T \mathcal{P}_{q,s}\, b_s(Y_s)\, \mathrm{d}s\right].$$

Proof: $\mathrm{d}_s(\mathcal{P}_{q,s} Z_s) = -\mathcal{P}_{q,s} b_s \mathrm{d}s + \mathcal{P}_{q,s} \Xi_s \mathrm{d}W_s$, a true martingale at fixed N.

Z = expected source on the unresolved scales, transported back, plus the boundary error. Compare $F_t = \mathbb{E}_t[F_T]$ in Lecture 4: here the source is the **defect**.

Three contributions

Bounded fields (two correction steps!):

$$\|Z_q^{\mathsf{b}}\| \le \int_q^\infty L_0 \frac{s}{q}\, C\gamma^{7/2} s^{-5/2} \mathrm{d}s = 2\, C L_0 \gamma^{7/2} q^{-3/2}, \qquad \frac{1}{\gamma}\int q^{1/2}\|Z^{\mathsf{b}}\|^2 \le C\gamma^6 q_0^{-3/2}.$$

Boundary: $q\|Z_q^{\mathsf{t}}\| \le L_0\, \mathbb{E}_q \|T Z_T\|$, contributing $\frac{2L_0^2}{\gamma q_0^{1/2}}\delta_{N,T,p}$, $\delta_{N,T,p} = \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}$.

Large fields: $A^{\mathsf{l}} = \int s^{7/2}\|b_s^{\mathsf{l}}\|^2 \le C\gamma^3 J_T$; Cauchy–Schwarz and Jensen,

$$\|Z_q^{\mathsf{l}}\|^2 \le 2L_0^2 q^{-5/2}\, \mathbb{E}_q A^{\mathsf{l}}, \qquad \left|\frac{1}{\gamma}\int q^{1/2}\|Z^{\mathsf{l}}\|^2\right|_{L^p} \le \frac{C\gamma^2}{q_0}\|J_T\|_{L^p}.$$

This is how the clipping error is controlled

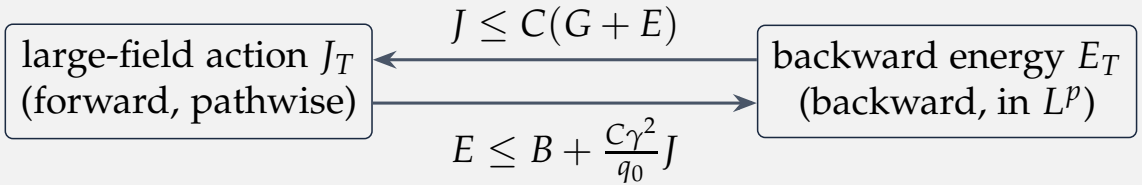
b^{l} is not small, but it enters the backward energy only through the large-field action J_T, with the small factor γ^2/q_0.

Closing the loop

Theorem. (cutoff-uniform energy estimate) *One $q_0 = q_0(\gamma)$: for all N, T, $p \geq 1$,*

$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \leq C_\gamma\,(1 + p^5 + \delta_{N,T,p}).$$

$x = \|E_T\|_p$, $y = \|J_T\|_p$: $y \leq A_0\,(C\gamma^3 p^5 + x)$, $x \leq B_0 + B_1\,\delta + \eta\,y$, $\eta = C\gamma^2/q_0 = C g_{q_0}^2$. Choose $A_0\eta \leq \frac{1}{2}$; $y < \infty$ at fixed N, T, absorb.



The smallness is in the **feedback**, independent of p: one q_0 for all p; the growth p^5 is inherited from G_T.

Consequences

Once the boundary term is removed (Lecture 7):

$$\sup_N \|E_\infty^N\|_{L^p} + \sup_N \|J_\infty^N\|_{L^p} \le C_\gamma\,(1+p^5), \qquad \sup_N \mathbb{E}\|Z_q^N\|_2^2 \le C_\gamma\, q^{-5/2},$$

and $\mathbb{E}\exp\{c\,(E_\infty + J_\infty)^{1/5}\} < \infty$.

- The approximate forecast differs from the exact one by $O(q^{-5/4})$ in $L^2(\mathbb{P}; L^2)$ (root mean square), uniformly in the lattice.
- $H_q Z_q = O(q^{-9/4})$ in $L^2(\mathbb{P}; L^2)$: up to an integrable drift, the forward equation is the explicit SDE $\mathrm{d}Y = b(Y)\mathrm{d}q + H(Y)\mathrm{d}W$.

Comparison with the variational method

Comparison with Boué–Dupuis (Lecture 4)

There: drift $u = -4\lambda \dot{Q}^{1/2} {:}X^3{:} + r$ ($\lambda \int {:}\varphi^4{:}$ of Lecture 4). Here: $\bar{\Gamma}$ does all such changes of variables at once in the forecast coordinate; Z plays the role of r. Coercivity of the quartic: **pathwise** in the forward equation; the remainder: **in expectation** through the backward one.

Barashkov–G. 2020